Fungal Growth Proliferates on Solar Panels: A Study of Three Sites Across North Carolina

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PreMiEr Precision Microbiome Engineering

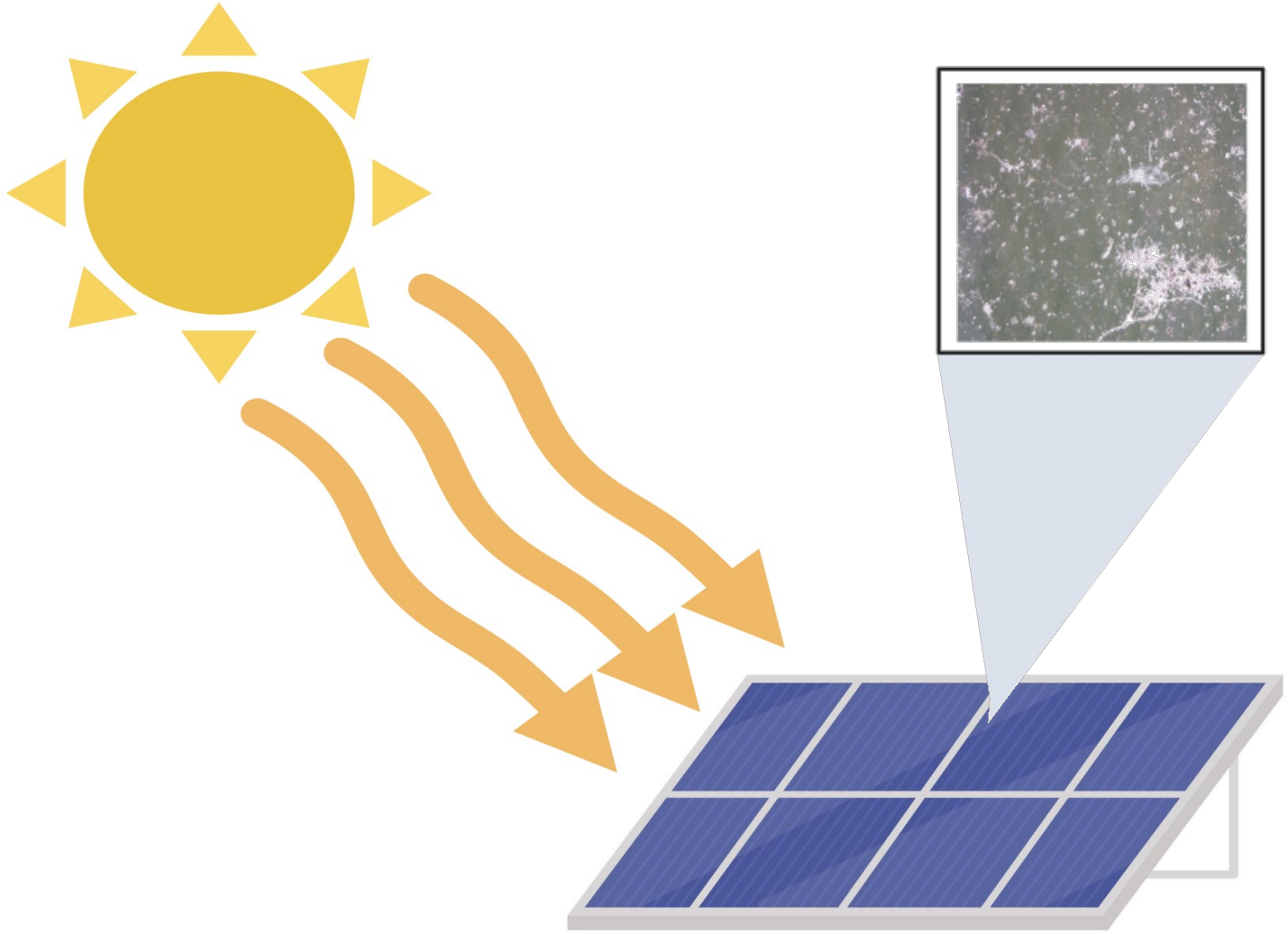
NSF Engineering Research Center

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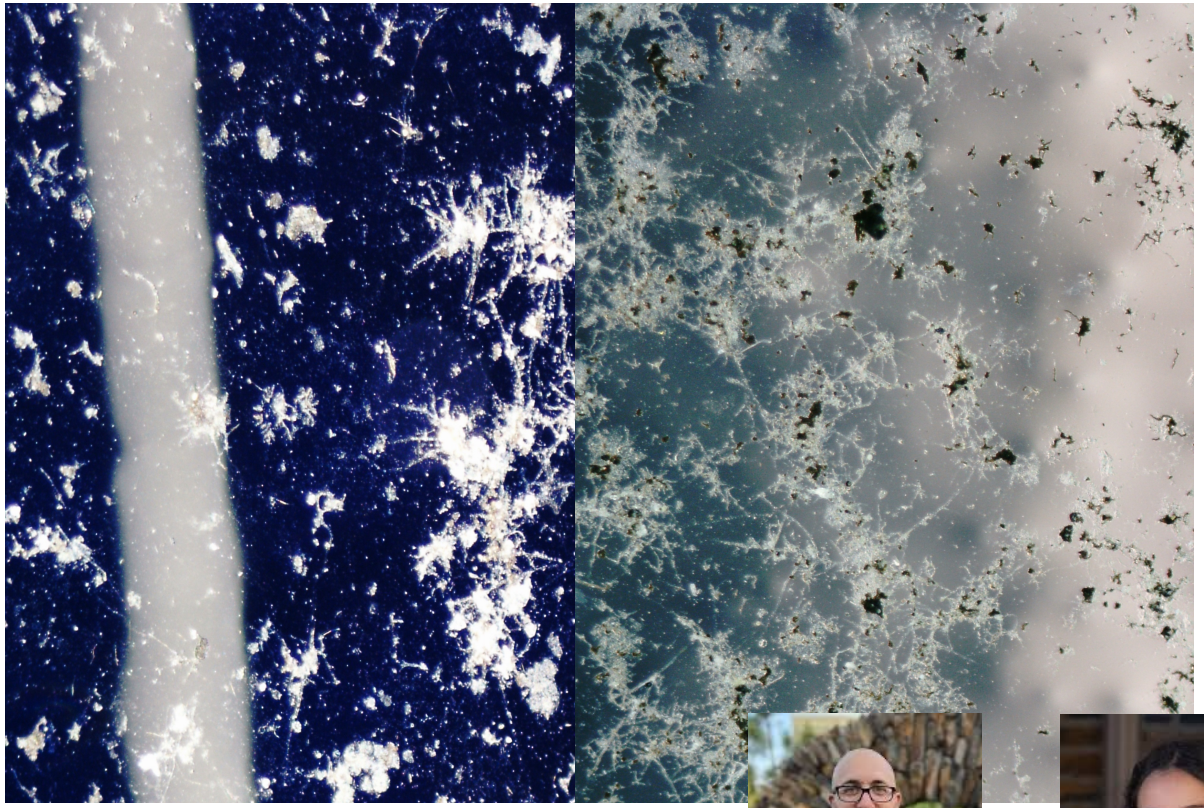
Solar Unsoiled

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Solar panels great renewable energy source



Working with Solar Unsoiled to enhance solar energy efficiency



fungal hyphae on solar panels

Image taken by Solar Unsoiled

Soiling appears to be primarily driven by **biotic factors** rather than the abiotic factors previously assumed

Solar Unsoiled



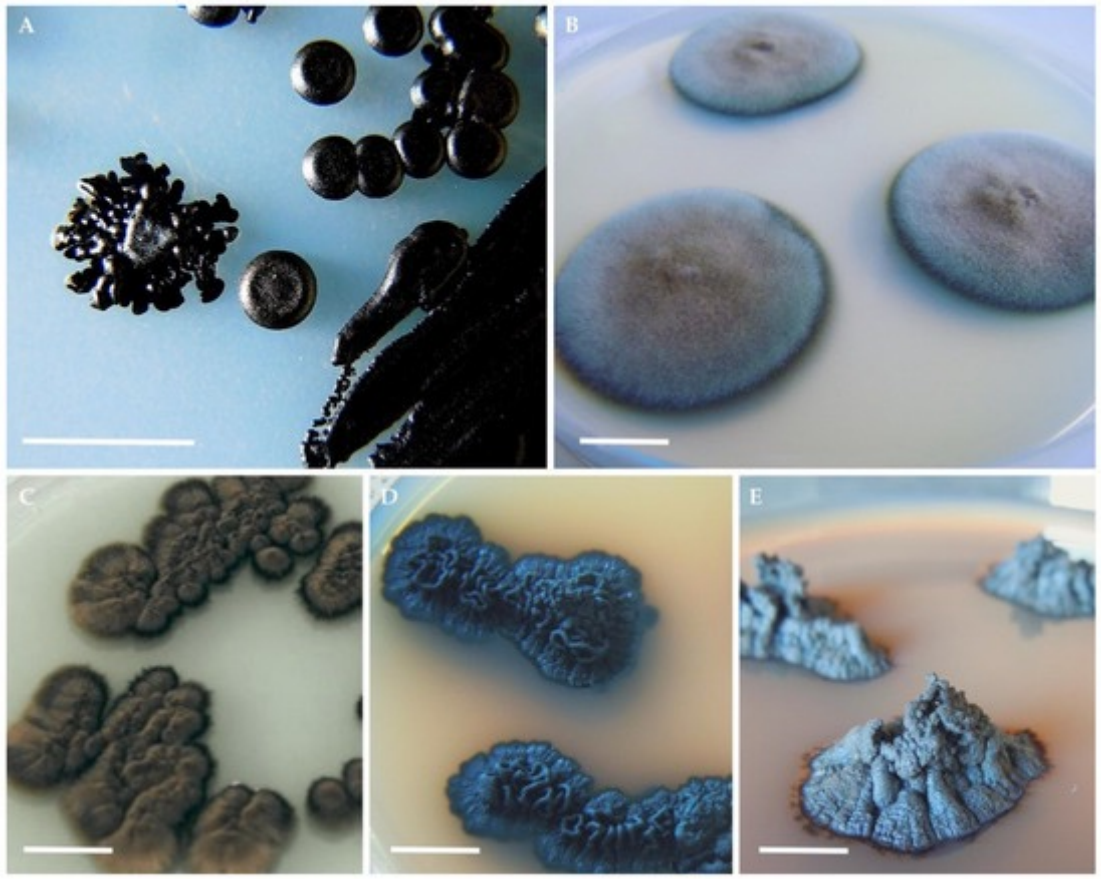
Mike Valerino



Alexandra Rivera

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Black fungi is a likely colonizer of solar panels



MDPI, 2022

Characteristics of black fungi that allow for survival on panels

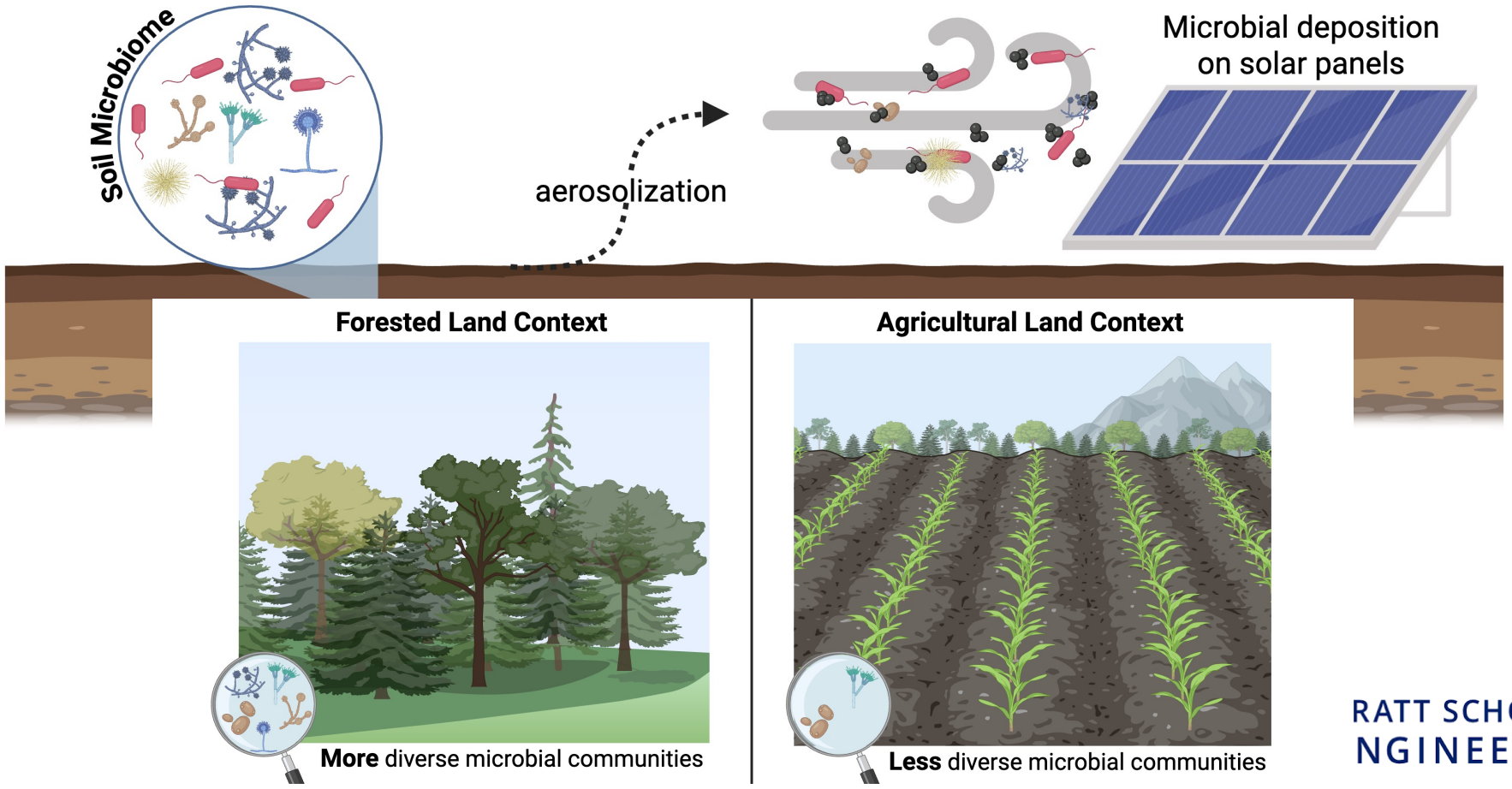
- Melanin production
- EPS production
- UV tolerant
- Ability to grow in low moisture

Journal of Fungi, 2023

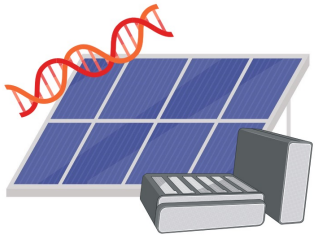
International Biodeterioration & Biodegradation, 2023

Life, 2020

Surrounding land use may shape fungal deposition on solar panels



Aim and hypotheses

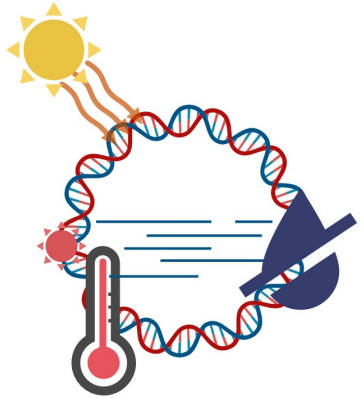


To characterize fungal colonization on solar panels, investigating the impact of surrounding land use to predict better cleaning strategies and cleaning frequency



1. Community Composition

The community will show dominance of black fungi, but will show site specific assemblages

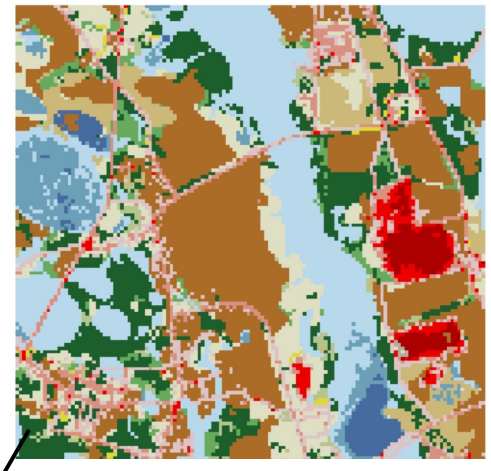


2. Functional Annotations

The community will show dominance of black fungi related survival genes, but will be conserved across sites

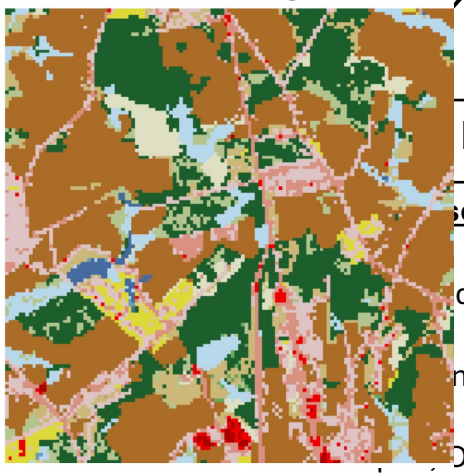
Land use differences in sampling sites across NC

Hope Mills NC

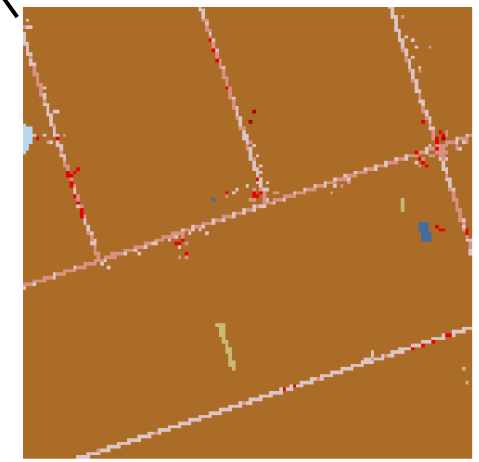




Laurinburg NC



Wilkinson NC





Hope Mills, N

Land Use Type	Per
Cultivated Crops	
Woody Wetlands	
Evergreen Forest	

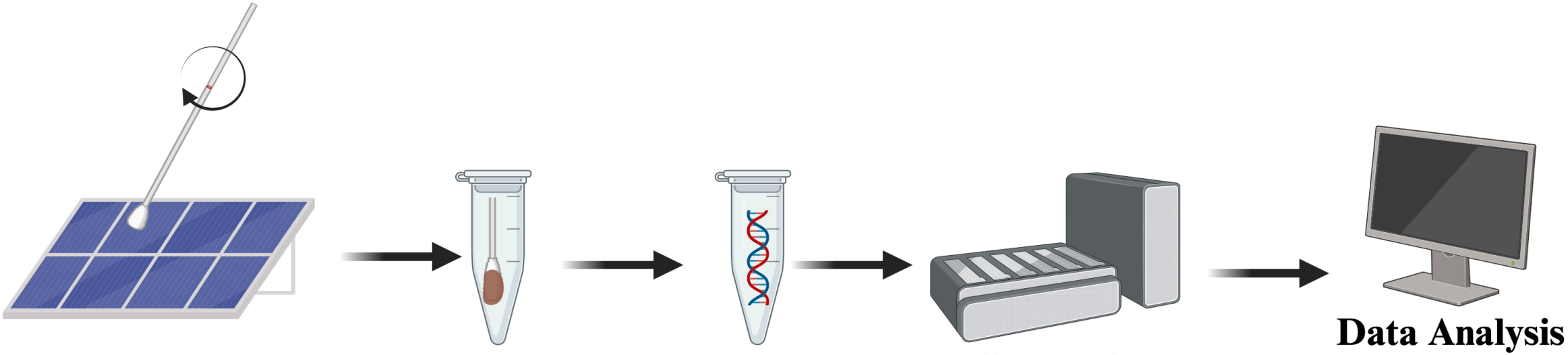
Laurinburg, NC

se Type	Percent Coverage
d Crops	39.66
n Forest	21.63
Open Space	10.57

NC

Percent Coverage
95.60
1.96
1.74

Methods for metagenomic analysis of solar panels



Sampling
PBS Soaked Swab

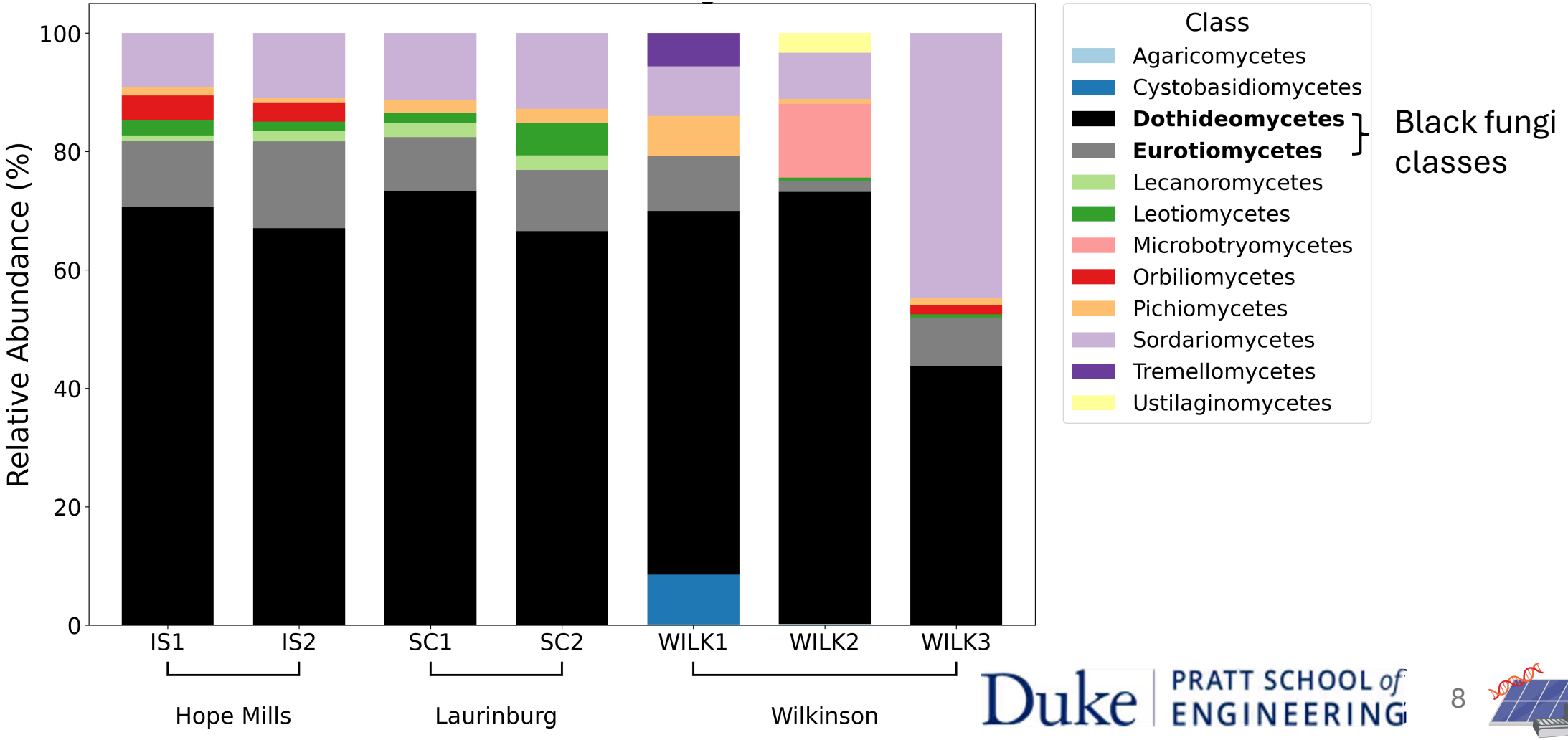
DNA Extraction
DNEasy Soil Power Pro Kit

Sequencing
PromethION

Data Analysis
Several online pipelines

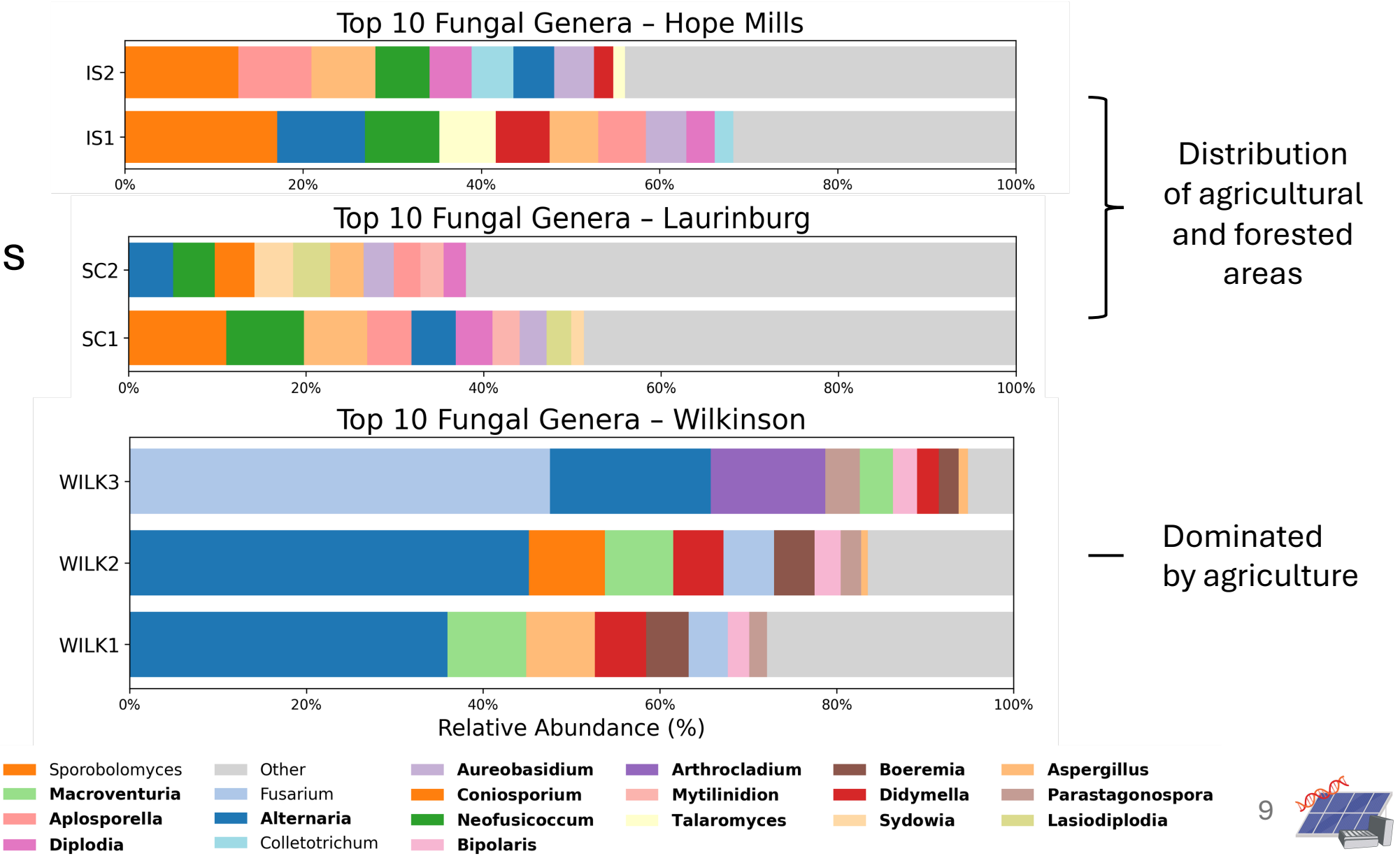
Prevalence of fungal classes

Solar panel communities are dominated by **black fungi**



Top 10 most abundant genera per site

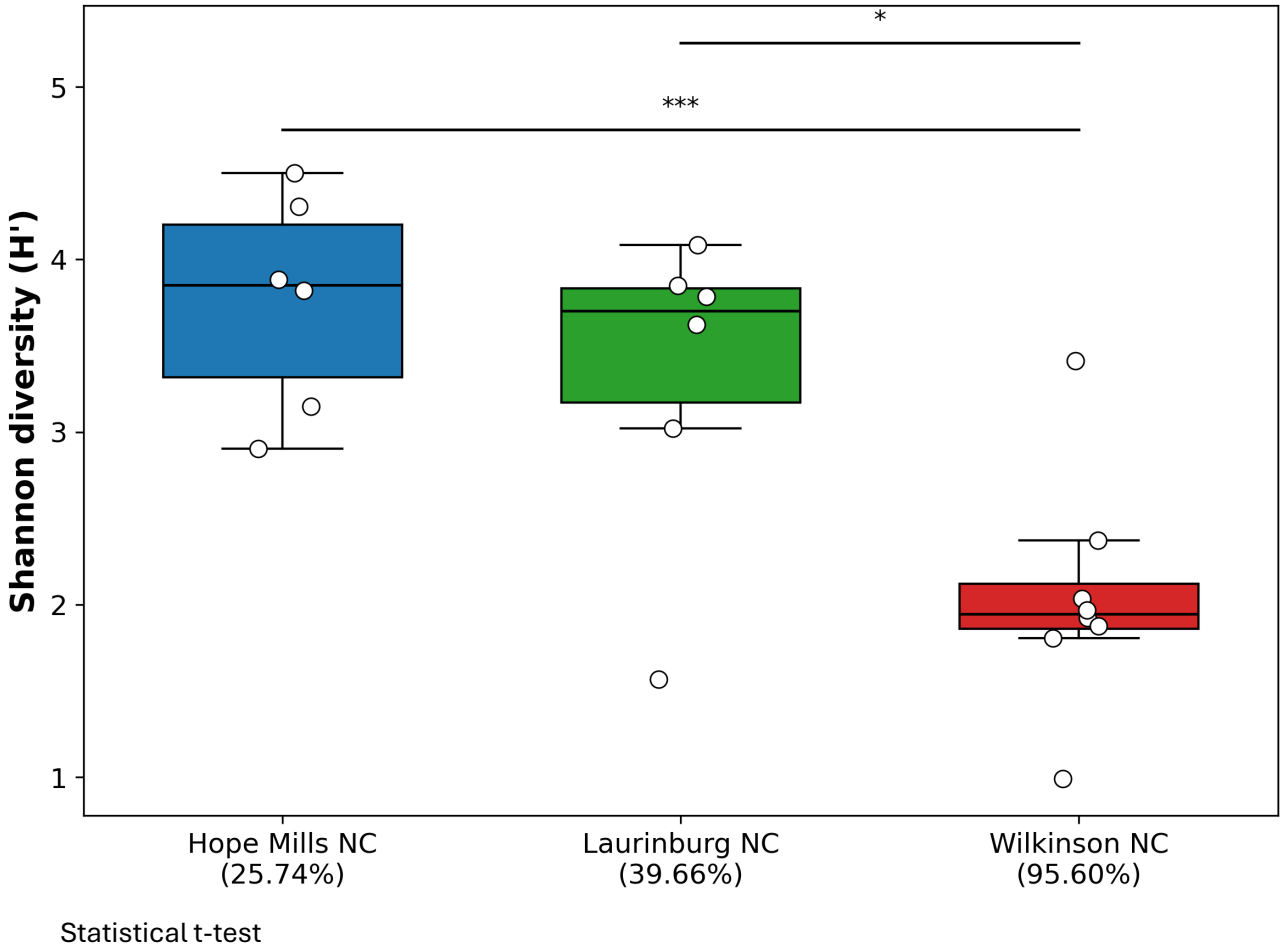
- Diversity is driven by **land use**



Distribution of agricultural and forested areas

Dominated by agriculture

Land use affects Shannon diversity



Statistical t-test

Measures the genus **abundance** and **evenness** within each sample

Sites with more surrounding agricultural land use showed **decreased Shannon Diversity**

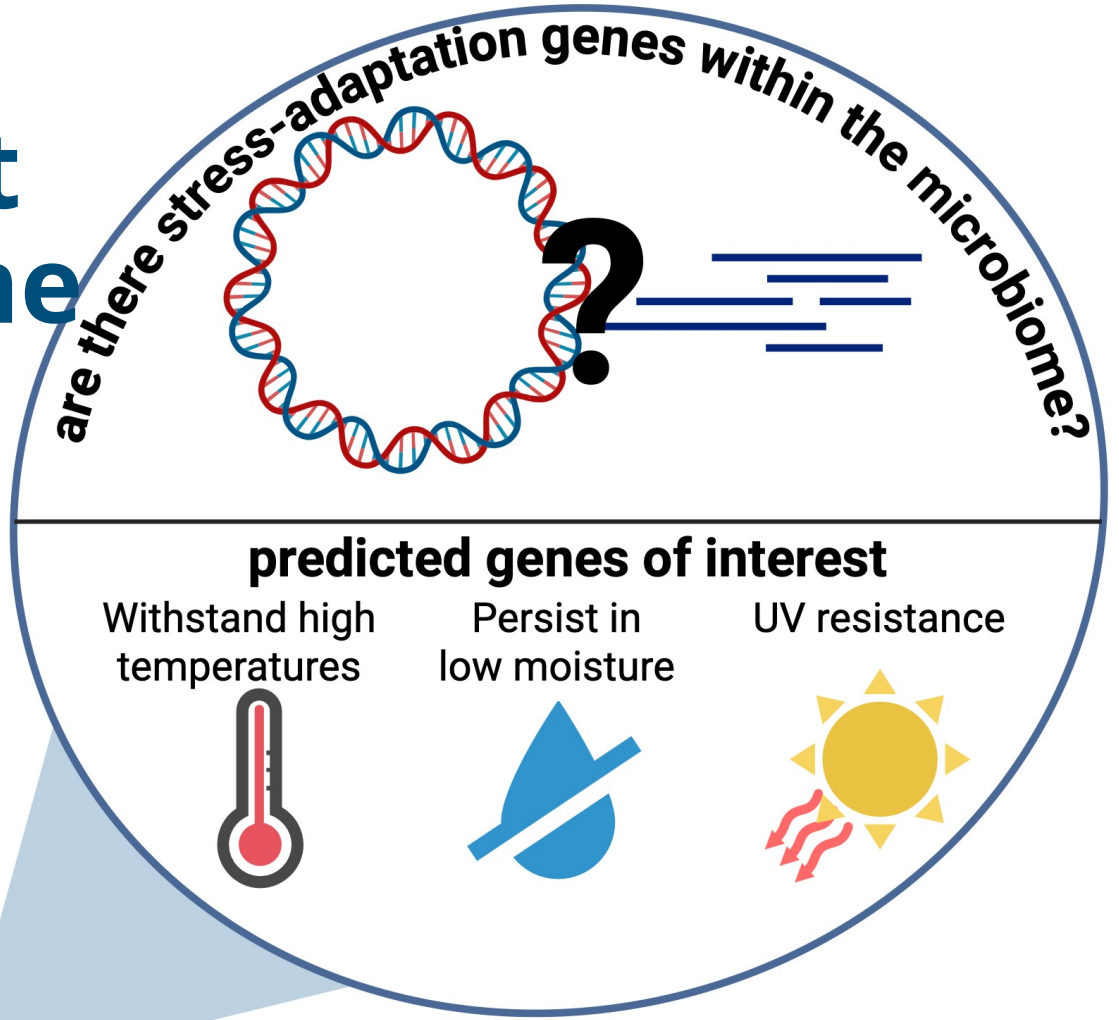
Shannon Diversity = more diverse samples

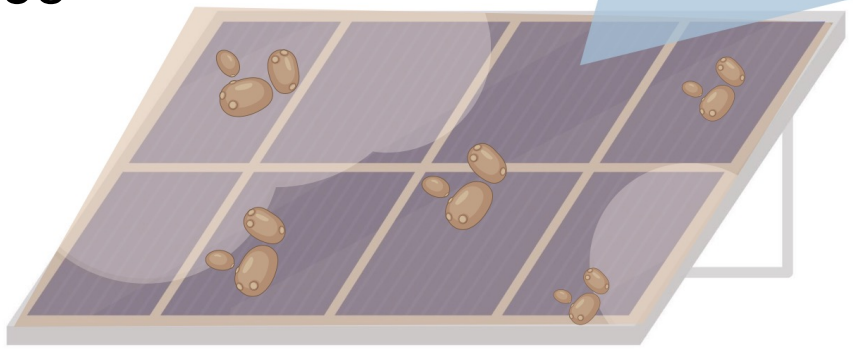
Searching for abundant genes in the microbiome

Using fastq files from Hope Mills, Laurinburg, and Wilkinson

Analyzing genes associated with:

- DNA repair
- Oxidative stress
- Heat resistance
- Drought resistance
- Transport
- Adhesion



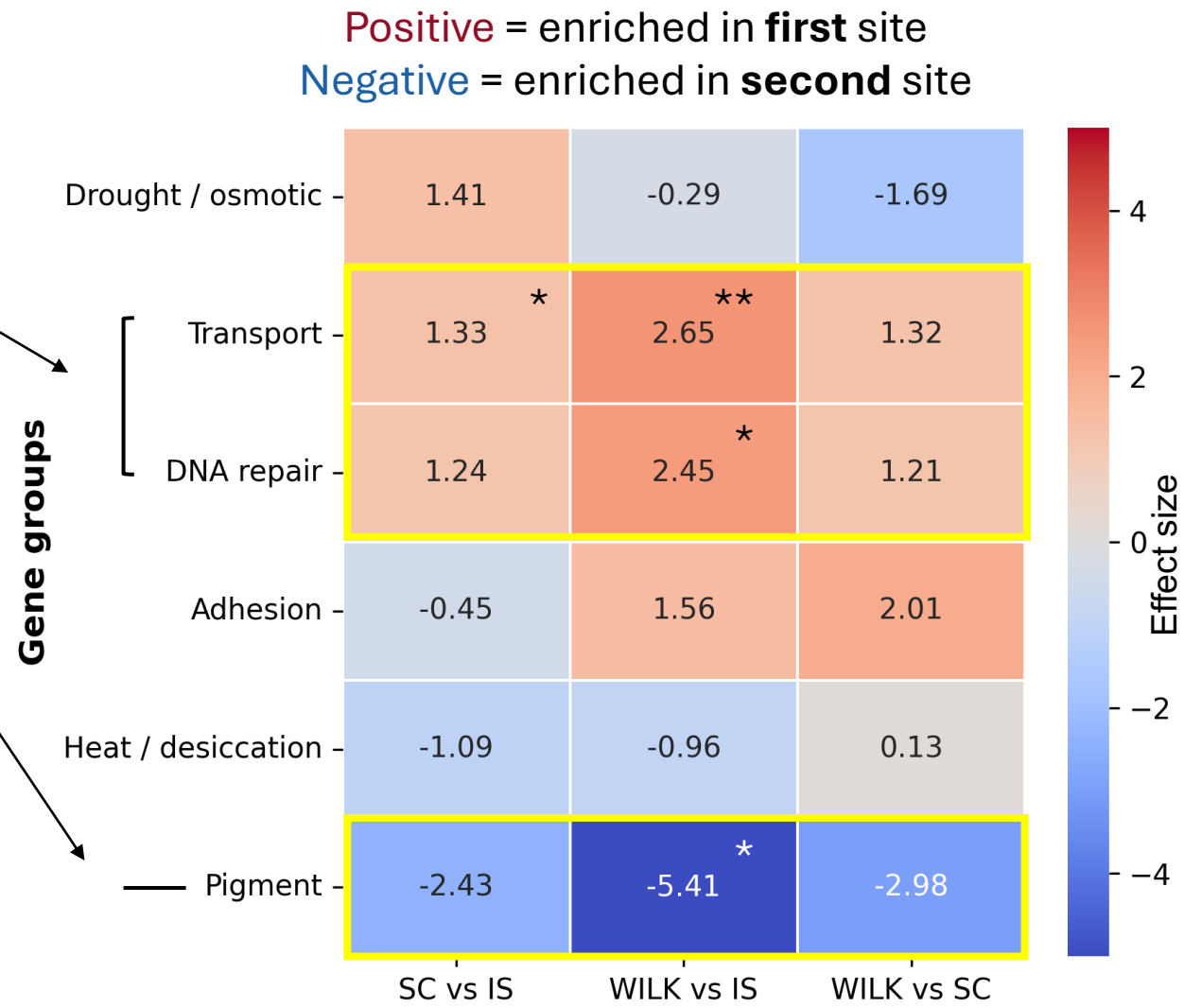


Site-Associated Shifts in Functional Gene Groups

Enriched at Laurinburg and Wilkinson relative to Hope Mills

Depleted in Laurinburg and Wilkinson relative to Hope Mills

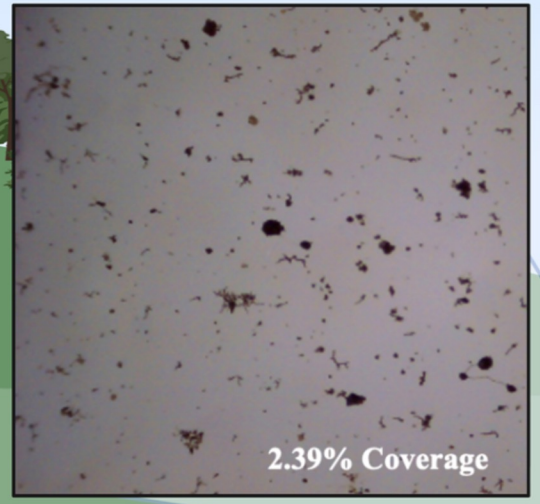
These patterns are consistent with land use trends, but have limited statistical power



Conclusions

- Black fungi are dominant colonizers on solar panels
- Surrounding agricultural land uses show a decrease in fungal diversity that colonized the solar panel surface
- Functional gene composition shows a directional shift across sites
- Other environmental factors play roles in microbial colonization

Next steps: develop a predictive model of solar panel soiling



1. Quantify soiling coverage from panel images

Thank you!!!

Gunsch Lab Members

Dr. Neil Gottel
Dr. Megan Hill
Dr. David Singleton
Amelia Foley
William Stiffler
Qing Dai
Andrew Wrench
Kayla Fericy

My Research Portfolio

SCAN ME

PreMiEr Precision Microbiome Engineering
NSF Engineering Research Center